Harpreet Singh

Embedded Firmware Engineer

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Summary

Senior embedded systems and firmware engineer with 13+ years building microcontroller products, board-support software, PCB bring-up tools, acquisition systems, industrial interfaces, and embedded Linux/Raspberry Pi applications. Strong C, Embedded C, Python, STM32/AVR/PIC/ESP32, SPI, I2C, UART, CAN, Ethernet/LWIP, SMPS, hardware debugging, and sensor-acquisition background, strengthened by current EEG/fNIRS and closed-loop BCI research work.

Core Skills

- C, Embedded C, Python, Bash, bare-metal firmware, FreeRTOS, Zephyr RTOS, Embedded Linux
- ARM Cortex, STM32, AVR, PIC, 8051, ESP32, Atmega328P, Atmega2560A, STM32F4
- SPI, I2C, UART, CAN, RS-485, Modbus, USB, Ethernet, MQTT, HTTP, WebSocket, LWIP
- PCB design, schematic design, board bring-up, JTAG, SWD, oscilloscopes, logic analyzers, SMPS, power management
- ADS1299, ADS8688, DAC8565, Raspberry Pi GPIO/SPI, live plotting, CSV logging, sensor validation

Professional Experience

Research Assistant, Core-Hub Neuroengineering Solutions | *University of Lethbridge | Apr 2025 - Present*

- Build embedded neuroengineering systems for EEG/fNIRS acquisition, SPI/GPIO device control, firmware validation, live plotting, and Python analysis workflows.
- Developed Raspberry Pi acquisition/control software using ADS1299, ADS8688, DAC8565, watchdog PWM, TDM MUX sequencing, threading locks, CSV logging, and hardware debug scripts.
- Integrated research-session software with CGX EEG, UDP-to-LSL markers, LabRecorder control, stream probing, XDF output, and session metadata.

Research Assistant | *University of Regina | Jan 2021 - Apr 2024*

- Researched uncertain-reasoning intrusion detection for DoS/DDoS detection using Bayesian Networks, Markov Networks, Zeek, Wireshark, feature engineering, and Python ML workflows.
- Published SECRYPT 2024 work on uncertain-reasoning IDS methods and presented findings to an international cybersecurity research audience.
- Built ML prototypes including a Bayesian liver-disorder diagnostic model with 85% accuracy and Punjabi BERT/NLP classifiers for 100K+ news articles with 92% classification accuracy.

Independent Embedded Systems Developer | *Self-Employed | Jun 2024 - Mar 2025*

- Designed a custom STM32F405RGT6 MicroPython development board, including multilayer PCB design, firmware flashing, sensor/actuator interfaces, power management, and hardware/software debugging.

Senior Software Engineer | *Planetcast Media Services Ltd. | Dec 2018 - Nov 2020*

- Led a 15-member team delivering embedded hardware and firmware for broadcast/media systems, including a network-controlled 10-channel video switcher and Ethernet temperature acquisition platform.
- Designed schematics and PCBs, selected components, collaborated on cabinet/mechanical integration, and developed C firmware with Ethernet, HTTP/LWIP, alarm, and control interfaces.

Embedded Software Engineer / R&D Engineer | *Spark Eighteen Pvt. Ltd.; Exicom Tele-systems Ltd. | Feb 2016 - Dec 2018*

- Developed STM32/AVR embedded systems for LiFi control, telecom power plants, DC interface cards, and automatic test equipment.
- Designed 15-20 W isolated SMPS sections with 78-81% measured efficiency and performed board bring-up across CAN, SPI Flash, I2C EEPROM, Ethernet/LWIP, and GUI test automation.

Selected Projects

BCT_8EEG_8FNIRS

- Developed Raspberry Pi acquisition/control software for ADS1299, ADS8688, DAC8565, shared SPI0, GPIO MUX/TDM sequencing, watchdog PWM, thread-safe SPI transfers, live plotting, and CSV logging.

BCI_Game_Loop_FPGA

- Defined embedded packet and transport contracts for FPGA-to-game control across Ethernet, Wi-Fi, UART, and SPI, including sequence numbers, class IDs, Q15 confidence values, and optional five-score payloads.

PyBoard / STM32 MicroPython Board

- Designed a custom STM32F405RGT6 development board with multilayer PCB, MicroPython firmware, sensor/actuator interfaces, and power-management debugging.

Publication

- Harpreet Singh et al., "An Uncertain Reasoning-Based Intrusion Detection System for DoS/DDoS Detection," SECRYPT 2024, ISBN 978-989-758-709-2, ISSN 2184-7711, pp. 771-776.
- Published thesis: Singh H. A robust intrusion detection system utilizing uncertain reasoning techniques in artificial intelligence. The University of Regina (Canada); 2024.

Education

- PhD in Computational Behavioural Neuroscience, University of Lethbridge - In progress
- M.Sc. Computer Science, University of Regina - 2024
- Bachelor's equivalent in Electronics & Telecommunication Engineering, AMIETE/IETE - 2012